

1.1 Atomic Structure

Question Paper

Course	CIEA Level Chemistry
Section	1. Physical Chemistry
Topic	1.1 Atomic Structure
Difficulty	Hard

Time allowed: 20

Score: /10

Percentage: /100

Question 1

The species Cl^- , K^+ and Ar are isoelectronic. This means that they have the same number of electrons.

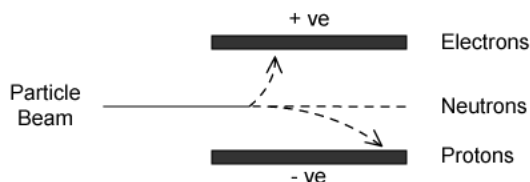
In which order do their radii decrease?

	Largest → Smallest		
A	K^+	Cl^-	Ar
B	Cl^-	Ar	K^+
C	K^+	Ar	Cl^-
D	Ar	K^+	Cl^-

[1 mark]

Question 2

The trajectories of subatomic particles shot through an electric field produced in a vacuum are shown in the diagram. The velocity is the same for all three particles.



Which statements are correct?

- 1** The proton and electron are deflected because they are charged particles.
- 2** The neutrons are undeflected as they have no charge.
- 3** The electron is deflected more than the proton because it has less mass.

A. 1, 2 and 3

B. 1 only

C. 2 and 3

D. 1 and 3

[1 mark]

Question 3

The atomic radius of the elements decreases moving across period 3. Which of the following statements explain this phenomenon?

- 1 Electrons shells are added moving across which increases the nuclear force of attraction.
- 2 The nuclear charge increases moving across due to increasing atomic number.
- 3 There is a greater force of attraction between the nucleus and the electrons.

A. 1 and 2

B. 1, 2 and 3

C. 2 and 3

D. 1 only

[1 mark]

Question 4

Element **X** is in period 2 and has the first seven ionisation energies in kJ mol^{-1} as shown.

1300	3380	5330	7460	11 010	13 320	71 200
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What is the electronic configuration of element **X**?

A. $1s^2 2s^2 2p^4$

B. $1s^2 2s^2 2p^2$

C. $1s^2 2s^2 2p^3$

D. $1s^2 2s^2 2p^6$

[1 mark]

Question 5

For the successive ionisation energies of oxygen, where would the highest jump be expected to occur?

- A. Second ionisation energy
- B. First ionisation energy
- C. Seventh ionisation energy
- D. Sixth ionisation energy

[1 mark]

Question 6

Which statement explains the similarity in chemical behaviour for isotopes of the same element?

- A. They may have the same electronic configuration, depending on the reaction.
- B. They have the same electronic configuration.
- C. Their mass numbers are different.
- D. They have different numbers of neutrons.

[1 mark]

Question 7

Which species produces a half-filled set of p orbitals on losing an electron?

- A. Li^+
- B. F
- C. N^+
- D. N^-

[1 mark]

Question 8

Which statement explains why electrons pair up in orbitals before occupying an orbital of a higher energy level?

- A. The energy from repulsion is greater than the jump to a higher orbital.
- B. Electrons occupy higher energy orbitals before lower energy orbitals.
- C. Electrons in lower orbitals have higher energies.
- D. The energy from repulsion is less than the jump to a higher orbital.

[1 mark]

Question 9

The first ionisation energy of beryllium is higher than the first ionisation energy of boron.

Which statement explains why?

- A. Boron has a full outer shell
- B. Boron has a larger atomic radius than beryllium
- C. Beryllium has a more stable electronic configuration
- D. The atomic number of beryllium is higher than boron

[1 mark]

Question 10

The electronic configurations of four different atoms are shown.

Which atom has the highest first ionisation energy?

- A. $1s^2 2s^2 2p^6 3s^2$
- B. $1s^2 2s^2 2p^4$
- C. $1s^2 2s^2 2p^6$
- D. $1s^2 2s^2$

[1 mark]

Model Answers: Hard

1

The correct answer is **B** because:

- All three species (K^+ , Cl^- and Ar) have the same number of electrons (18) and hence the same number of electron orbitals.
- Therefore the size of each species is determined by the **proton number**.
- The species with the most protons will have the **most positive** nuclear charge, therefore exerting the biggest “pull” on the negatively charged electrons.
- K^+ has 19 protons, Ar has 18 and Cl^- has 17, so K^+ will have the smallest radii whereas Cl^- will have the largest.

A, C & D are incorrect as these options do not place the species in the correct order of decreasing radii.

2

The correct answer is **A** because:

- The upper plate is positively charged and the lower plate is negatively charged, so protons and electrons are deflected to the plate which is opposite their charge.
- Neutrons pass straight through **unaffected** as they are neutral.
- The electrons are deflected **more** because they have a **much smaller mass** than the protons but carry the same but opposite charge.

B, C & D are incorrect as all three statements are correct.

3

The correct answer is **C** because:

- Moving across the period the **atomic number** increases by 1 for each element, hence the nuclear charge becomes greater.
- The electrons are added but to the **same energy level**, hence the increase in **shielding** is negligible.

A, B & D are incorrect as statements 2 and 3 are correct. Statement 1 is incorrect as the electron shells are added moving down a group, not across a period.

4

The correct answer is **A** because:

- Removing electrons from higher energy orbitals requires less energy than removing electrons the lower energy orbitals.
- There is a steady increase in the first four ionisation energies, indicating that the first four electrons all came from the **same energy level**.
- The **large jump** between the fourth and fifth ionisation energies (from 7450 to 11 000) indicates an electron has been removed from the next energy level down which is 2s.
- The relatively small jump to 13 300 corresponds to the second of the 2s electrons.
- Another large jump occurs (from 13 300 to 71 000) which corresponds to removing an electron from the 1s energy level.

B, C & D are incorrect as these electronic configurations do not match the ionisation energies of element X.

5

The correct answer is **C** because:

- The electronic configuration of oxygen is $1s^2 2s^2 2p^4$.
- The largest jump (difference in energies) occurs between removing the 6th electron, from the 2s energy level, and the 7th electron is being removed from the 1s energy level.

- The first shell (1s) is much closer to the nucleus, so electrons in this shell are much more **strongly attracted** to the nucleus and don't experience **shielding**.
- This means there is a greater force of attraction between the nucleus and the seventh electron to be removed, so more energy is needed to overcome this force.

A, B & D are incorrect as these ionisation energies involve electrons being removed from the second shell where the **distance** to the nucleus is greater and there is more **shielding** taking place from the inner shell of electrons.

6

The correct answer is **B** because:

- Isotopes are atoms of the same element with the same **atomic number** but a different number of neutrons.
- Their electronic configuration is always the same since they have the same numbers of protons and electrons.
- The mass number is different due to the different numbers of uncharged neutrons.
- Therefore they have similar chemical properties.

A is incorrect as their electronic configurations are the same.

C & D are incorrect as the mass number affects their physical properties only, such as boiling point or melting point.

7

The correct answer is **D** because:

- N^- loses an electron to produce N which has the electronic configuration $1s^2 2s^2 2p^3$.
- The p orbital can hold 6 electrons, hence at 3 electrons it is half-filled.

A is incorrect as the atomic number of lithium is 3 hence it doesn't have any p orbitals.

B is incorrect as the electronic configuration of F^+ would be $1s^2 2s^2 2p^4$.

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C is incorrect as the electronic configuration of N^+ would be $1s^2 2s^2 2p^2$.

8

The correct answer is **D** because:

- The **inter-electron repulsion** between electrons in an orbital is less than the energy required to jump to a higher orbital.
- The electrons thus pair up in their respective orbitals first before occupying orbitals of higher energy.

A is incorrect as the repulsion between electrons in the same orbital is less than the energy required to jump to a higher orbital.

B is incorrect as electrons occupy orbitals of lower energy first.

C is incorrect as the energy of the orbitals increases the further away they are from the nucleus.

9

The correct answer is **C** because:

- The electronic configuration of beryllium is $1s^2 2s^2$ and therefore the first ionisation requires removing an electron from a **full 2s subshell**.
- The electronic configuration of boron is $1s^2 2s^2 2p^1$ and therefore the first electron will be removed from the 2p subshell, which requires less energy as it is not full and is placed **further away** from the nucleus than the 2s subshell.

A is incorrect as the outer shell of boron is not full.

B is incorrect as atomic radius **decreases** moving across the table.

D is incorrect as the atomic number of boron is one more than beryllium.

10

The correct answer is **C** because:

- The electronic configuration of option C indicates that the element is **neon**.
- First ionisation energies generally increase moving across the periodic table from left to right.
- Neon is a noble gas and has a **full outer shell** which makes it very stable and difficult to ionise.
- More energy is required to remove an electron from a complete shell than one with unpaired electrons or an incomplete shell.

A, B & D are incorrect as these atoms have unfilled outer shells hence their ionisation energies are not as high as an atom with a full outer shell.